ECE1786 progress report - ProdMatch

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1 Introduction

In today's digital marketplace, users face a daunting challenge: sifting through an overwhelming array of products to find ones that genuinely meet their needs. While traditional recommendation systems rely on keyword-based searches and collaborative filtering, these methods often fail to understand nuanced user preferences and context. The rise of multi-modal systems offers a transformative opportunity to bridge this gap.

ProdMatch aims to revolutionize product discovery by leveraging LMMs to interpret complex user queries and product descriptions, delivering recommendations that go beyond superficial keyword matches. This project seeks to transform e-commerce by addressing limitations in existing systems and enhancing the user experience with precise, personalized, and intent-driven product suggestions. By combining textual and visual data, ProdMatch envisions a solution that reflects true user intent, making product search intuitive and effective.

2 Input/Output

Our model accepts two types of inputs: a product-based input and a user query. Both inputs are processed to extract detailed features.

2.1 Type 1: Product-Based Input

The input consists of a textual prompt and a reference to images representing a product. The prompt provides the product's title, price, and a request for feature extraction based on the image and title.

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Input Prompt

- **Product Image:** Reference to the images of a dining table that contain the dimension and the colour of the table.
- **Product Title:** " HOMCOM Dining table High Gloss White"
- **Price:** "$468.70."
- **Task:** "Extract features based on the image and title."

Output Response

The output is a response generated by the model, which analyzes the provided information and returns the extracted features in a structured format. For example:

- **Color:** "White"
- **Dimensions:** "L: 31.5 inches, W: 31.5 inches, H: 29.92 inches."
- **Price:** "$468.70."

2.2 Type 2: User Query

The input is a user query, where the user provides specific product requirements such as colour, dimensions, and price. For example:

- **Input User Query:** "Can I find a black table that's 29.53 inches in height, 39.37 inches in length, and 19.69 inches in depth, priced at $250?"

Output Response

The output for this type of input is similar to the product-based input, where the model extracts and provides detailed product information matching the query. For example:

- **Color:** "Black."
- **Dimensions:** "L: 39.37 inches, W: 19.69 inches, H: 29.53 inches."
- **Price:** "$250."

Current Development and Final Goal

The project's ultimate goal is to return direct links to products that meet the user's requirements but right now the outputs are only images of five tables that match the user's prompt. These images serve as a visual representation of the closest matches and help in refining the final functionality to include links.

3 Data

Our dataset is divided into two main parts: a product database and user queries.

3.1 Product Database

We collected photos and data of dining tables and desks from Amazon, assigning a unique ID to each product. Images, dimensions (length, width, height), colour, descriptions, and their corresponding labels (length, width, height, colour, and price) were stored. All information, including features and a structured prompt guiding feature extraction (analyzing images and titles for colour, dimensions, and price), was saved in a JSON file.

3.2 User Queries

The second part of the data is user queries. These are users' inputs which specify their requirements, including features such as colour, dimensions, and price.

4 Baseline Model or Comparison Method

We are doing a class 2 project, evaluating three pre-trained multimodal models: InternVL2-2B, InternVL2-4B, and InternVL2-8B. Our manually created dataset, sourced from Amazon, includes product images, descriptions, prices, colours, and sizes. After fine-tuning the models on the training set, we test them on the same test set and compare prediction accuracies, calculated as the ratio of correctly predicted labels to total labels.

5 Architecture

Our ProdMatch system is a multimodal platform that integrates user input and product data, enabling precise and personalized product recommendations. The model working process can be split into 2 main steps: user prompt processing and product matching.

5.1 User Query Processing

Input: Users provide natural language descriptions of their needs (e.g., "I'm looking for a red dining table under $100").
Processing: The query is parsed using the finetuned multimodal model InternVL2-8B, which extracts relevant features (e.g., colour: red, type: dining table, price range: <$100). Then the extracted features are converted into the same structured format used for database entries, ensuring compatibility during matching.

5.2 Product Matching and Recommendation

Input: The structured features from the user query.
Processing: Custom matching functions compute how closely each product aligns with the user's described needs.
Output: A ranked list of top-matching products is generated and presented to the user.
A key innovation here is multimodal embeddings are used to align visual and textual features, ensuring that products with matching visual and descriptive attributes are prioritized.

5.3 Current Progress

We have finished building the database manually (which will be expanded after), fine-tuned three multimodal models InternVL2-2B, InternVL2-4B, and InternVL2-8B, and compared their test accuracies.

6 Result

We used three multimodal models: InternVL2-2b, InternVL2-4b, and InternVL2-8b. When testing on the user query dataset, all three models have a perfect feature extraction ability(100%). When testing on the product features dataset, the test accuracies are 0.1, 0.7, and 0.8 respectively. For the 2b model, the colour recognition was mostly inaccurate, and there was confusion in the recognition of size, with the model often reversing the length and width. For the 4b model, the colour recognition is better, but there are still mistakes shown when recognizing sizes. The 8b model shows a good recognition ability when facing relatively simple pictures but still struggles to extract all sizes correctly if the image is tremendously complex.

7 Discussion

Our final choice is the InternVL2-8b model. When reaching a 100% accuracy in extracting features from user prompts and 80% accuracy in recognizing features like colour, size, and price from product descriptions and images, we believe the model can complete the task better if our training set is larger. So next step our focus will move to the completion of our dataset.

	product accuracy	user query accuracy
vl2b	0.3	0.4
vl4b	0.7	0.8
vl8b	0.8	1.0

Figure 1: Prediction Result

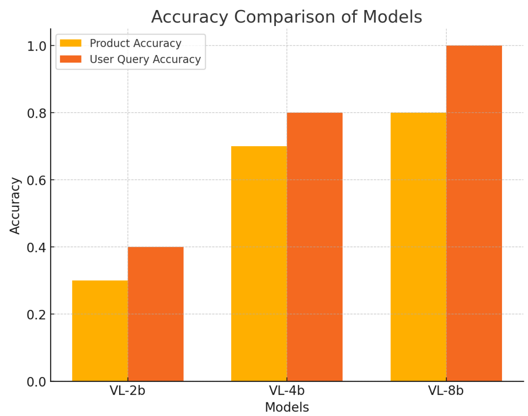

Figure 2: Accuracy Comparison

8 Team Work and Progress

8.1 Project Progress

We completed the dataset collection by November 10 and the model training by November 19, as planned. Based on feedback from the TAs, we have decided to remove the design of the UI component, as it was deemed unnecessary. Instead, we will focus on refining the model application and enhancing dataset scalability.

8.2 Work Distribution

Task	Status	Responsibility
User Query Dataset	Completed (expansion ongoing)	Chen Wang, Jiaguan Tang, Bocheng Zhang
Product Info Dataset	Completed (expansion ongoing)	Chen Wang, Jiaguan Tang, Yue Lan, Bocheng Zhang
Model Training	Completed	Jiaguan Tang
Model Evaluation	Completed	Bocheng Zhang
Progress Report	Completed	Chen Wang, Yue Lan
Model Application	Ongoing	Jiaguan Tang, Bocheng Zhang

Table 1: Project Milestones and Status